Intro to Deep Learning for Earth Observation

From Spectral Data to Land-Cover Maps

Build one small model. Change one thing.
Explain what happens.

هَاكَاثُون الشَّباب العَرَبِيّ
THE ARAB YOUTH HACKATHON

Understand
The learning workflow

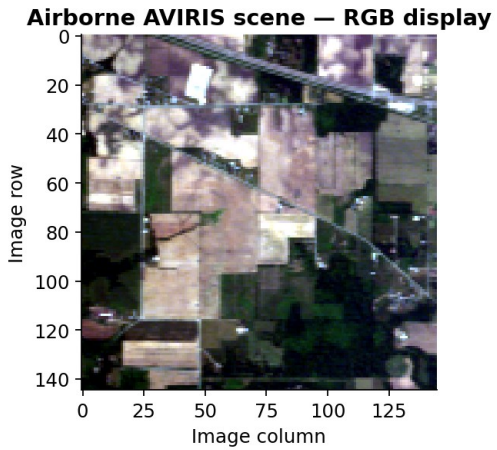
Experiment
Hands-on notebook

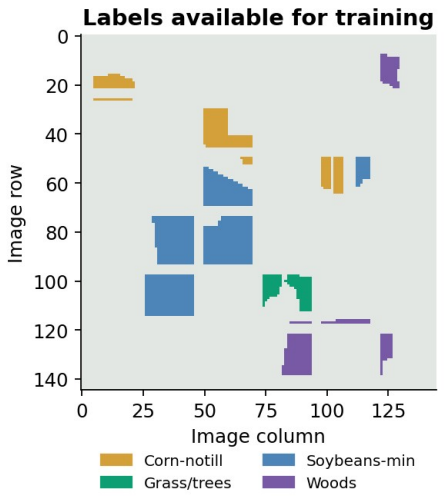
Discuss
Results and limitations

Ask
Your project questions

One pixel. One class. One useful experiment.

A small land-cover task that we can inspect from end to end.





OUR REAL TEACHING DATASET

Indian Pines / AVIRIS (1992)

145 × 145 pixels · 220 bands

4 labelled land-cover classes

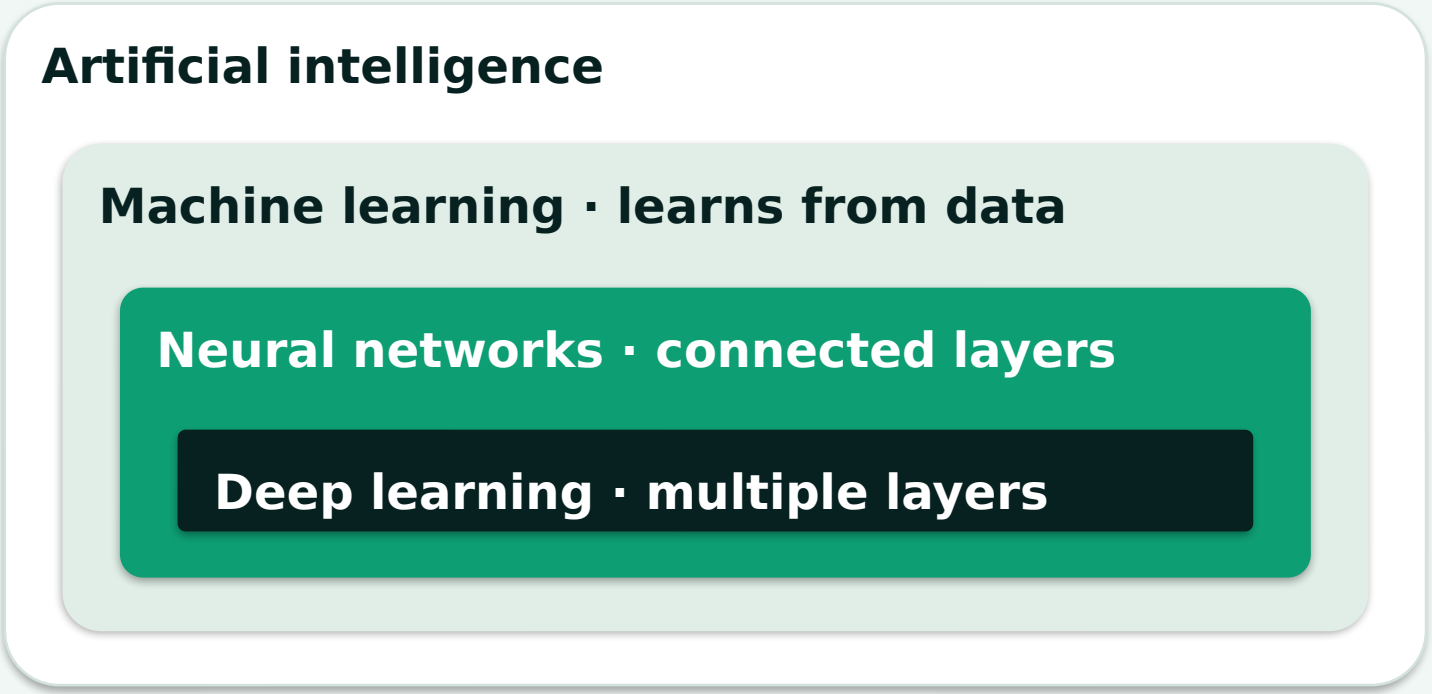
Input: a pixel's spectral values

Output: a predicted class

The map is a prediction — not automatically a reliable product.

Where does deep learning fit?

A practical family tree of today's neural-network methods.



WHAT MAKES IT DEEP?

Layers learn useful representations from data.

Not only chatbots.
Not always a huge model.
No universal layer-count cutoff.

This is a family tree — not a ranking of quality.

A useful model needs more than a model

Five connected parts — illustrated by our supervised land-cover task.

01

Question

Land-cover class.
Score: macro-F1.
Inspect errors.

02

Data + labels

Pixel spectra.
Four classes.
Separate areas.

03

Model

A small network.
Inputs → layers
→ class scores.

04

Learning

Loss + optimiser.
Learning rate.
Epochs.

05

Checks

Validation picks.
Freeze the choice.
Test checks it.

Compute and software make this possible. They do not prove it works.

If the labels or the test are wrong, a bigger model will not fix the evidence.

Choose a method. Not a winner in advance.

Start with a useful baseline. Compare on the same task and held-out data.

APPROACH	WHEN IT MAY FIT	EFFORT AND CAUTION
Rules / thresholds	A clear measurable signal	Easy to inspect; conditions can change.
Classical ML	Structured or spectral features	Start simple; labels and testing still matter.
Adapt pretrained DL	A model suited to your inputs	May save training; check task/sensor mismatch.
Neural net from scratch	Task and resources justify it	Training cost depends on model size and data.

COMPLEXITY

Build, run and maintain it.

QUALITY

Measured performance on the task.

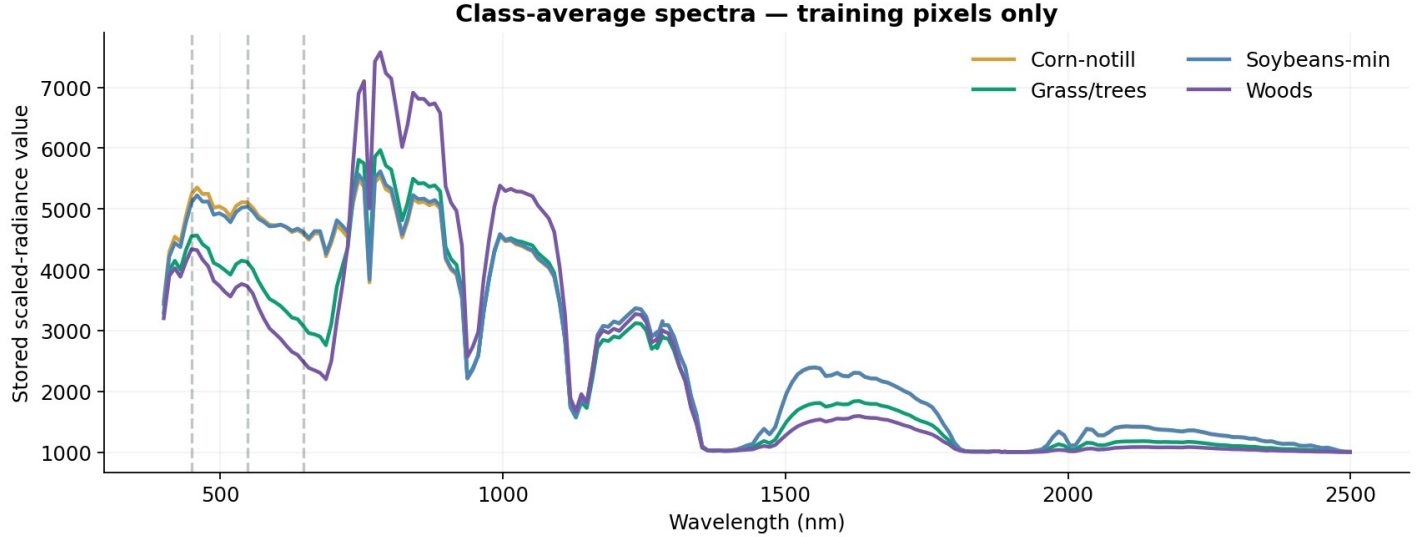
RELIABILITY

Evidence it holds where needed.

More complex does not automatically mean more accurate — or more trustworthy.

An EO image is more than a picture

At each location, the sensor measures a vector of values.



Different classes can have different spectral patterns.

More bands may help.
They may also be redundant, noisy or harder to learn from.

Predict first: which classes might be hard to separate?

How does the neural network learn?

Weights turn spectral inputs into a score for each class.

Spectral inputs
One pixel's values

→

Hidden: 32
Weighted sums + ReLU

→

Hidden: 16
Weighted sums + ReLU

→

4 class scores
Choose the largest

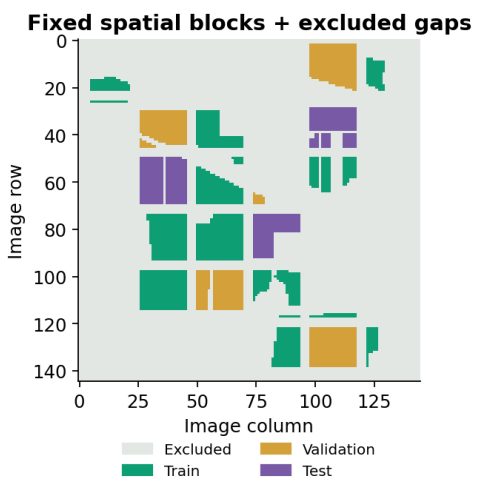
Predict → Measure loss → Update weights

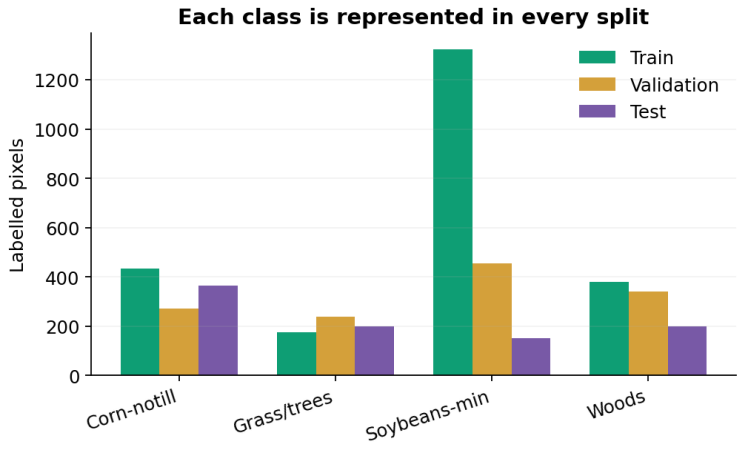
Training: predict → measure loss → update. Inference: predict only.

Epoch = one pass through training data. Learning rate = update size.

Training is not testing

Nearby pixels are similar. Our split keeps spatial areas separate.





TRAIN

Learn weights and scaling.

VALIDATION

Compare experiment settings.

TEST

Evaluate the frozen choice once.

One scene is still one scene — this does not prove transfer to a new region.

Open the notebook. Make a first prediction.

Start here → data → split → reference model.

REFERENCE CONFIGURATION

```
BAND_SET = "rgb"
EPOCHS = 5
LEARNING_RATE = 0.003
TRAIN_FRACTION = 1.0
```

01 Run the setup cells.

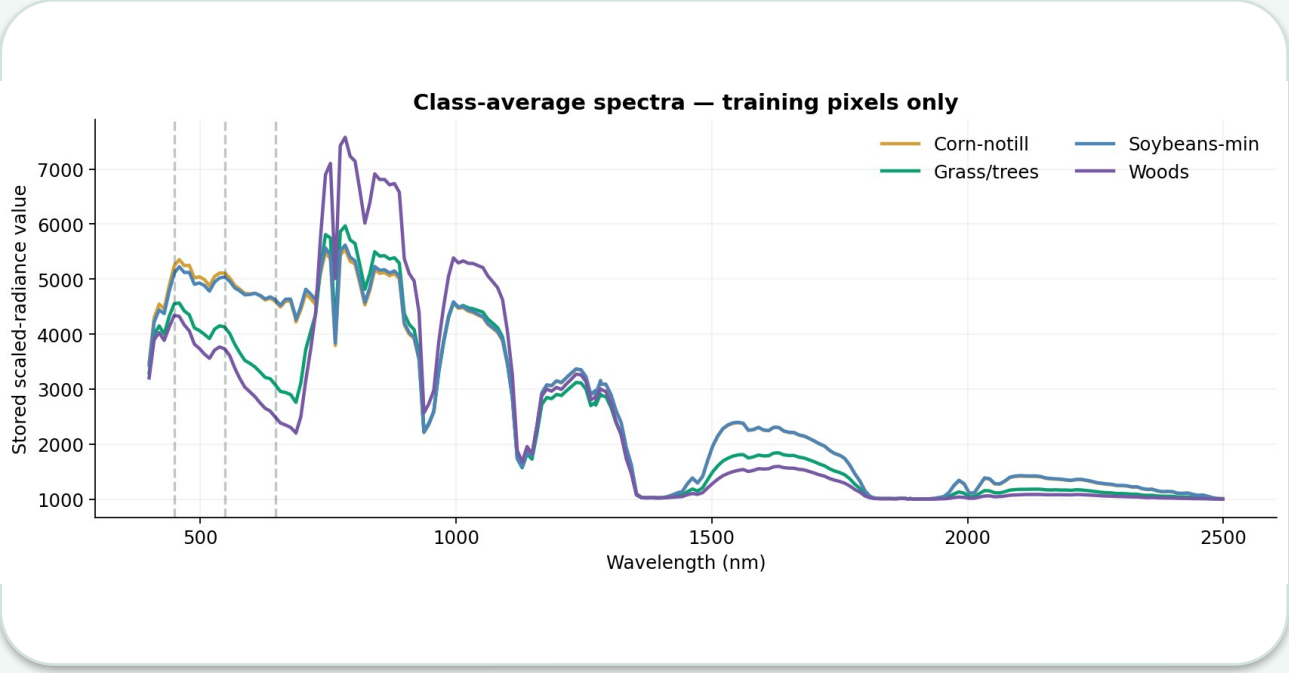
02 Inspect data and split.

03 Train the reference model.

Before you run: what would a useful result look like?

Experiment 1 — Do more bands help?

Change BAND_SET. Keep the model, seed and split fixed.



PREDICT

Which classes may benefit?

CHANGE + RUN

Selected bands ↔ full band set

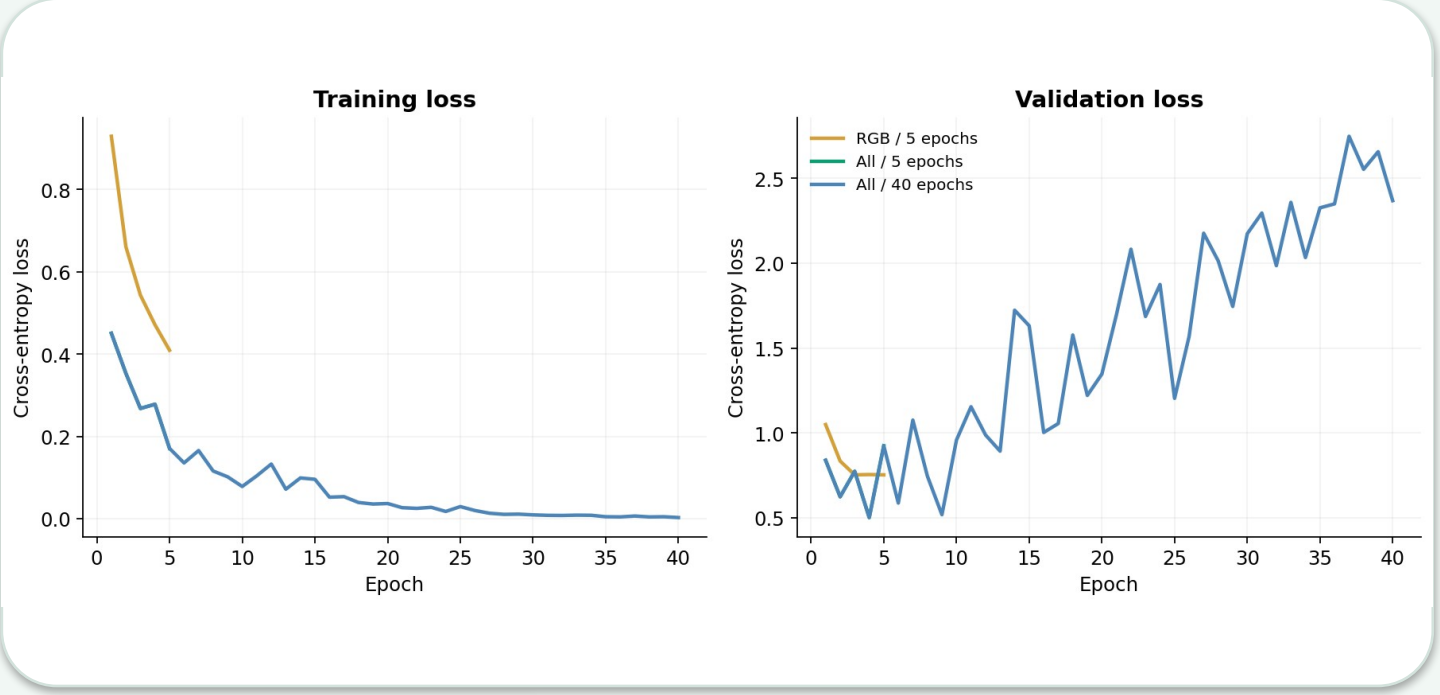
COMPARE + EXPLAIN

Validation score and class errors

Same hidden layers; more inputs also mean more trainable weights.

Experiment 2 — Does longer training help?

Keep BAND_SET = 'all'. Compare 5 and 40 epochs.



WATCH THE GAP

Training loss can improve while validation does not.

F1 can rise while loss worsens: confident mistakes cost more.

Where would you stop — and which evidence supports your choice?

Explore one more thing.

Optional experiments — complete the main two first.

OPTION A

Learning rate

Control the size of weight updates.

Do the curves improve steadily, move slowly or become unstable?

OPTION B

Training examples

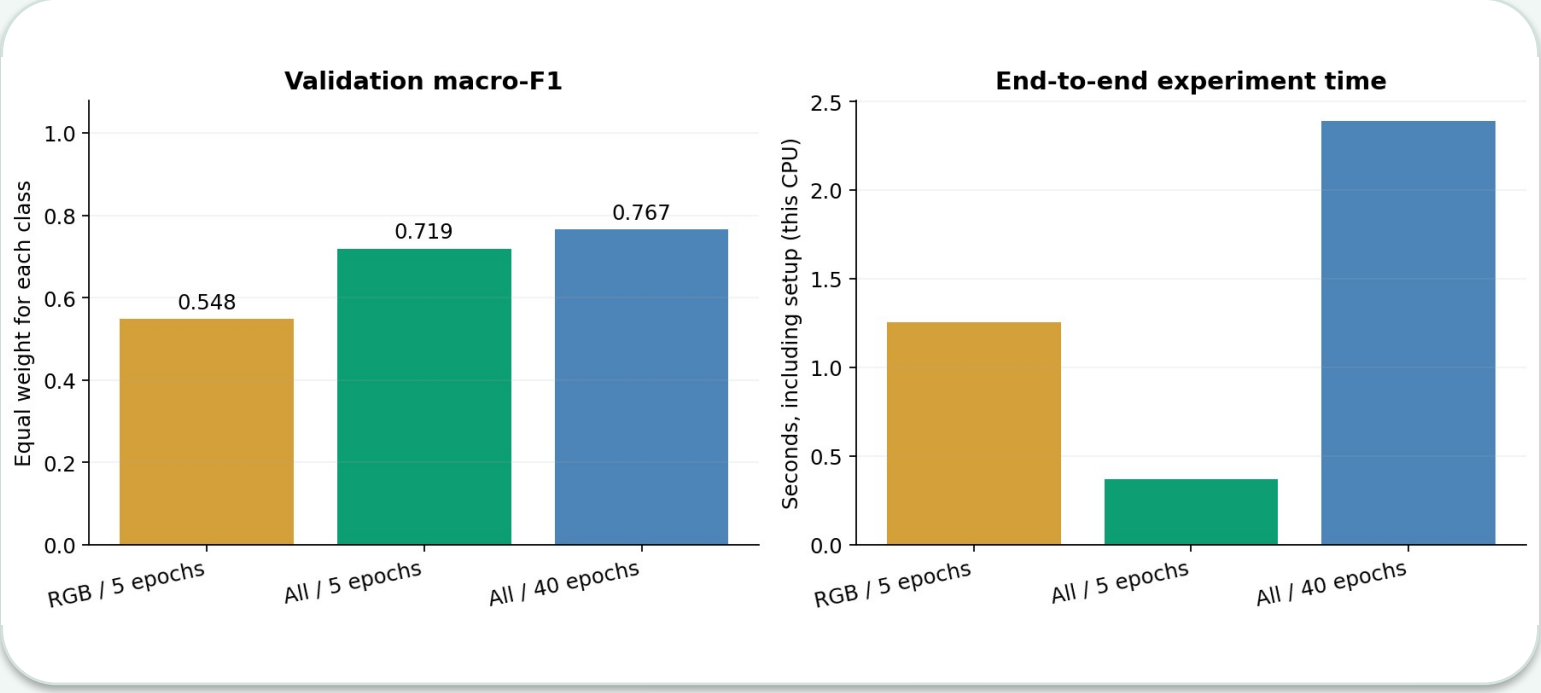
Use fewer labelled training pixels.

How much labelled data does this small experiment need?

Predict → change one factor → run → compare → explain.

Read the results, not just the best score

These are the measured instructor runs — your run is another observation.



What changed?

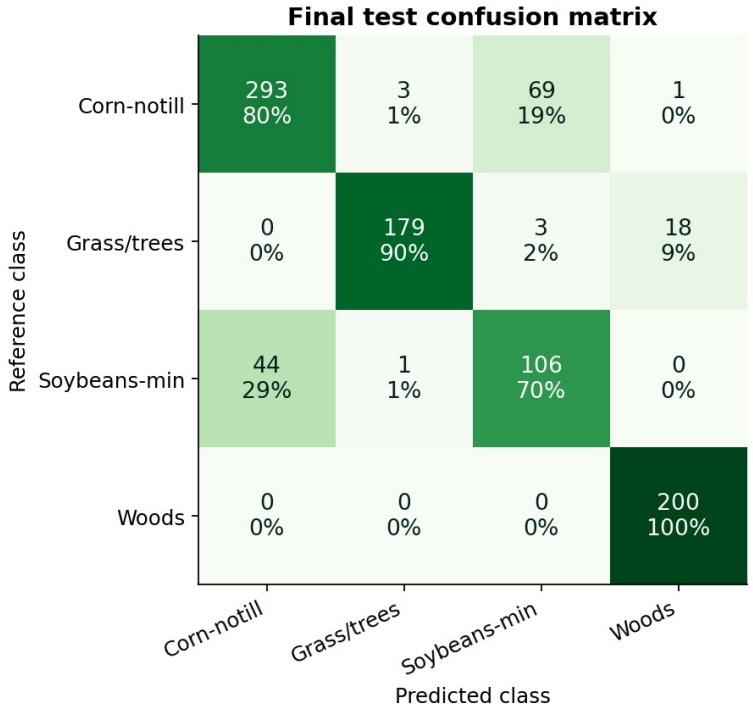
Did it help enough?

What did it cost?

Choose a configuration using validation. Then freeze it.

Which mistakes matter?

Final held-out test: evaluate the choice we already made.



TEST MACRO-F1

84.2%

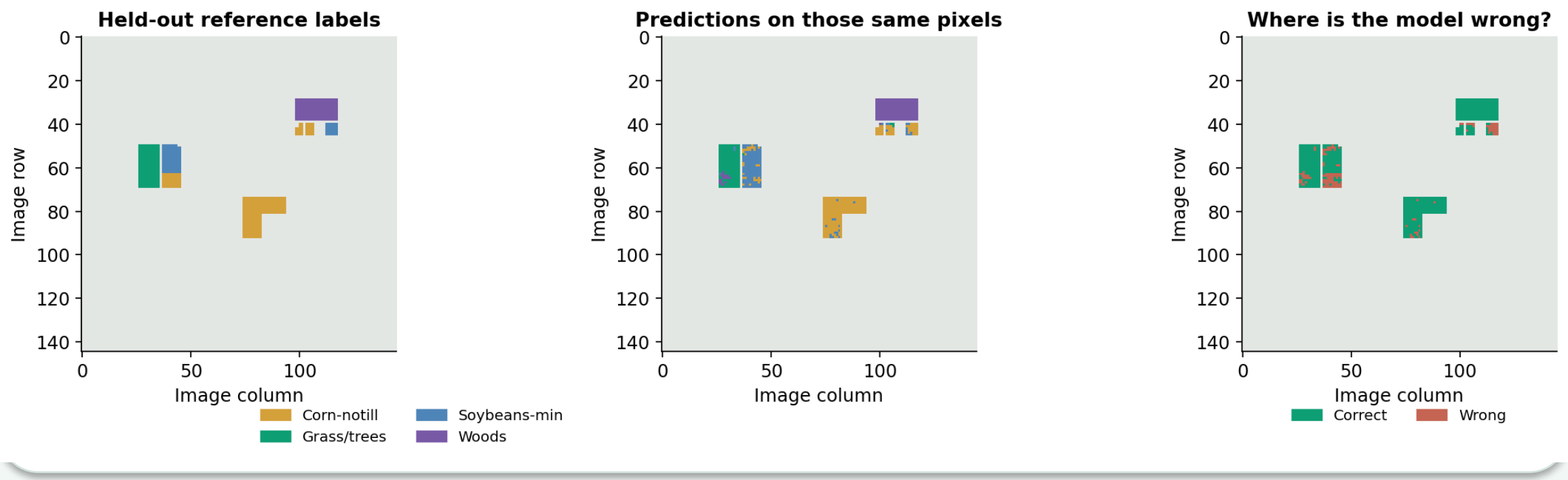
Pixel accuracy: 84.8%

Inspect the confusion matrix.
Name one important error.

A good average can still hide a bad error for your use case.

From pixel predictions to a land-cover map

Look for spatial patterns, boundaries and suspiciously confident mistakes.



Final-test pixels only · image coordinates · one airborne scene

What could break outside this notebook?

A useful PoC makes its assumptions visible.

01	**A new place or date**	Different surfaces, seasons, lighting or atmosphere.
02	**A different sensor**	Different bands, resolution and preprocessing.
03	**Incomplete labels**	Missing classes, uncertain boundaries, class imbalance.
04	**A different decision**	The same error may have a very different real cost.

What new evidence would make you trust the model more?

Keep it small. Make the evidence clear.

01 Start with the decision, data and labels.

02 Change one thing and record what happened.

03 Use validation to choose; test to check.

04 A map and a high score are the start of the discussion.

Questions?

DATA & PROVENANCE

Baumgardner, Biehl & Landgrebe (2015), Purdue. DOI: 10.4231/R7RX991C.

CC0 1.0 · Full sources in SOURCES.md and speaker notes.